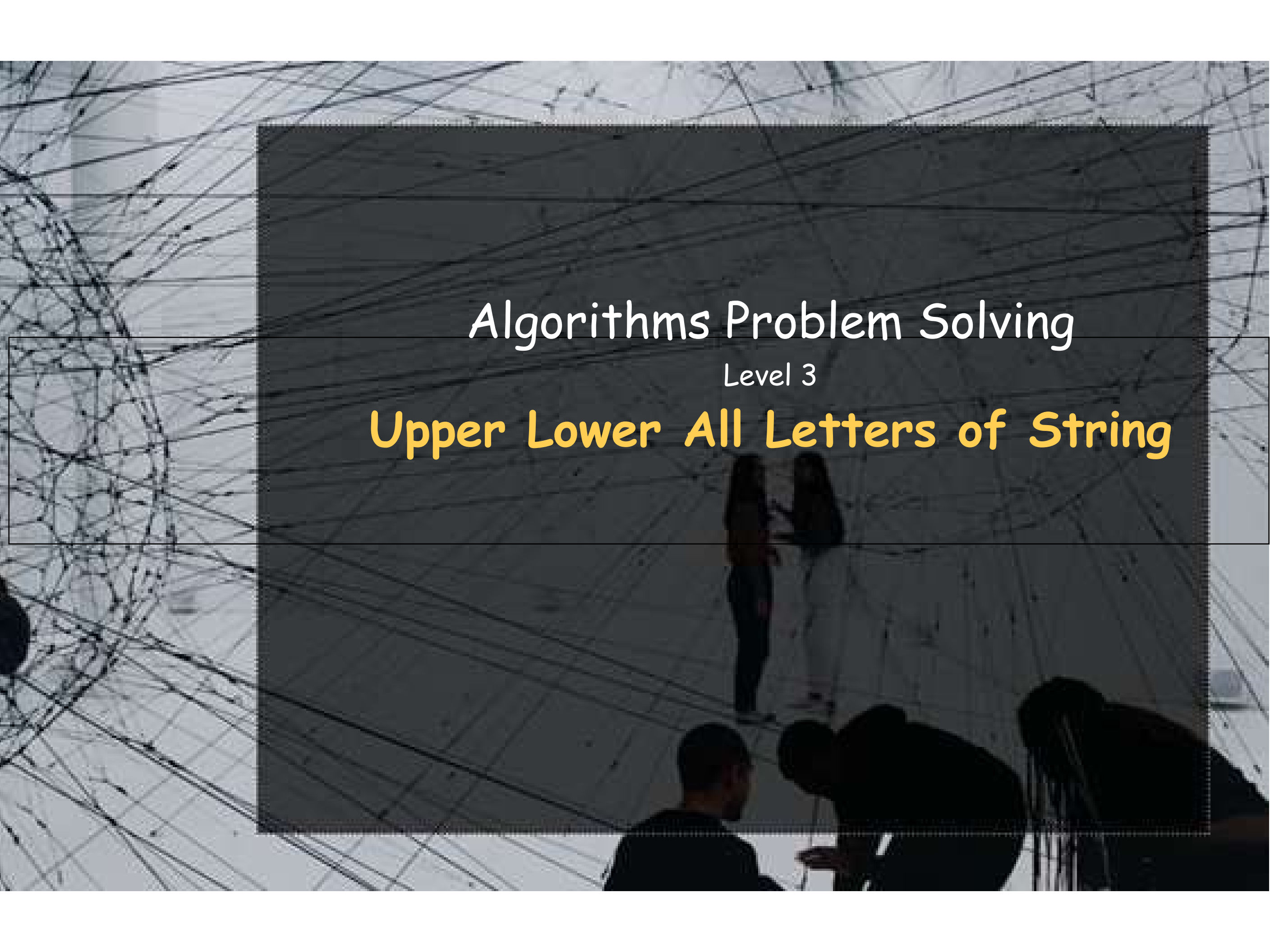


Algorithms Problem Solving

Level 3

Upper Lower All Letters of String



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Definition

Complete String Case Conversion

 Program provides functions for converting every applicable character in a string to uppercase or lowercase.

 These operations are useful for normalization and case-insensitive text processing.

Explanation

Uppercase Transformation

 ``UpperAllString()`` traverses the entire string and applies ``toupper()`` to each character.

 Lowercase letters are converted to uppercase.

Lowercase Transformation

 ``LowerAllString()`` traverses the entire string and applies ``tolower()`` to each character.

 Uppercase letters are converted to lowercase.

Character Preservation

 Numbers, spaces, and punctuation generally remain unchanged because they have no uppercase or lowercase form in the basic character classification used here.

Procedures

1. Define `UpperAllString()`.
2. Define `LowerAllString()`.
3. Read a complete string using `getline()`.
4. Apply both transformations.
5. Print the resulting strings.

💡 Example

Input:

Sara Alterkawi

➡ Result

String after Upper:

SARA ALTERKAWI

String after Lower:

sara alterkawi

Note

- ◆ The functions can modify the original string or return a transformed copy.
- ◆ Case normalization is commonly used before case-insensitive comparisons.
- ◆ Basic ``<cctype>`` functions are primarily character-based and do not provide full Unicode case conversion.

⊘ Warning

- ◆ ``toupper()`` and ``tolower()`` return an ``int``, so character handling should be performed correctly.
- ◆ The complete string must be traversed to transform every character.
- ◆ Passing by reference is required when the function is intended to modify the caller's string directly.



Comparison

▨ `UpperAllString()`

- ✓ Converts applicable letters to uppercase.
- ✓ Processes the complete string.
- ✓ Leaves non-letter characters unchanged.

▤ `LowerAllString()`

- ✓ Converts applicable letters to lowercase.
- ✓ Processes the complete string.
- ✓ Leaves non-letter characters unchanged.

✦✦ Summary

- ✓ Traverses every character in a string.
- ✓ Provides complete uppercase conversion.
- ✓ Provides complete lowercase conversion.
- ✓ Demonstrates reusable case-normalization utilities.

Thank you 😊